Numerical Study of the 1D Advection–Diffusion Equation with Parameter Optimization

Abstract

This work presents a numerical solution of the one-dimensional Advection–Diffusion equation using the Finite Difference Method. A simple optimization approach is applied to estimate the physical parameters of the system.

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1. Introduction

The Advection–Diffusion equation describes transport phenomena in physics and engineering. It combines transport (advection) and spreading (diffusion) effects.

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2. Governing Equation

$$du/dt + v\, du/dx = D\, d^2u/dx^2$$

Where:

- $u(x,t)$: transported quantity
- v: velocity
- D: diffusion coefficient

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3. Numerical Method

The equation is solved using:

- Finite Difference Method (FDM)
- Explicit time stepping scheme
- Discretization in space and time

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4. Optimization Strategy

The parameters (v, D) are selected using a grid search method that minimizes a cost function defined as:

$J = S\ u(x,T)^2$

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5. Results

The model returns:

- Optimal velocity (v)
- Optimal diffusion coefficient (D)
- Final concentration profile

The results show stable numerical behavior and physically consistent diffusion patterns.

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6. Conclusion

A simple but effective numerical framework was developed to solve and optimize the Advection–Diffusion system. The method can be extended to more complex inverse problems and real-world applications.